

SCD Dbase Sorter - Project Documentation

Document: DEPLOYMENT_SECURITY.md

Deployment Hardening Guide

This guide provides instructions for hardening the deployment of the SCD Dbase Sorter application.

1. Nginx Reverse Proxy with HTTPS

It is strongly recommended to run the Streamlit dashboard behind an Nginx reverse proxy. This provides SSL/TLS termination, better performance, and an additional security layer.

Nginx Configuration Example

```
```nginx
```

```
server {
```

```
ssl_certificate /etc/letsencrypt/live/yourdomain.com/fullchain.pem;
```

#### Strong SSL settings

# SCD Dbase Sorter - Project Documentation

location / {

proxy,

## Streamlit-specific settings

proxy,

## 2. SSL/TLS Certificate (Let's Encrypt)

Use `certbot` to obtain and automatically renew certificates:

```
```bash
```

sudo

3. Firewall Configuration (UFW)

Restrict incoming traffic to only necessary ports:

```
```bash
```

sudo

# SCD Dbase Sorter - Project Documentation

## 4. Intrusion Prevention (Fail2Ban)

Install and configure `fail2ban` to protect against brute-force attacks:

```
```bash
```

```
sudo d
```

Create `/etc/fail2ban/jail.local`:

```
```ini
```

```
[sshd]
```

```
[nginx-http-auth]
```

```
enable
```

## 5. Environment Variable Management

Do NOT hardcode secrets (like SMTP passwords or Master Keys) in the code. Use a `.env` file and load them using `python-dotenv`.

Example `.env` file:

```
```env
```

SCD Dbase Sorter - Project Documentation

Ensure ``.env`` is added to ``.gitignore``.

6. Directory Permissions

Ensure the ``data/`` directory has restricted permissions:

```
```bash
```

**Dat**